

Mahmoud A. Aboulata

Personal Information

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Links	LinkedIn GitHub

Interests and Goals

I have a mix of project experience in Data Engineering, Machine Learning, and pipeline systems design. My interests are to apply end-to-end data infrastructure and automated analytics stacks to solve complex, real-world problems.

Skills & Tech Stack

○ Programming & Query Languages	- Python · SQL · Bash · Java · C++ · html/Js/Css
○ Data Engineering & Infrastructure	- Apache Kafka · Apache Spark · Apache Airflow · dbt Debezium · Delta Lake · PostgreSQL · Docker · Kubernetes · Terraform · AWS
○ ML/AI & Data Science	- Scikit-learn · XGBoost · TensorFlow · PyTorch · HuggingFace SHAP · MLflow · Evidently AI · FastAPI - Pandas · NumPy · Matplotlib · Seaborn · Plotly Statsmodels · Prophet · Power BI · Tableau
○ Advanced Methods	- Feature Engineering · Time Series Analysis · A/B Testing Causal Inference · Survival Analysis · NLP · Anomaly Detection
○ Data Governance & Compliance	- PII Handling · Model Auditing · Fairness Analysis Egyptian PDPL · GDPR Article 22 · Audit Logging
○ Cloud & DevOps	- AWS (S3 · Glue · Redshift · Kinesis · Lambda) GitHub Actions · CI/CD · Linux

Education

2024	- BSc. (Hons) Informatics & Computer science (Artificial Intelligence Major) GPA: 3.1/4 Graduation Year: 2028
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Experience & portfolio - [GitHub](#)

Independent (Freelance) & Personal projects:

All projects are either through independent contracts, freelancing or personal exercising | **Started 2026**

Automated Data Platform

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| • Project | - Egyptian Banking & Financial Inclusion Analysis |
| • Ingestion & Processing | - Built an automated Python and API data extraction pipeline from scratch with input validation and exception handling. |
| • Orchestration & Database | - Managed processing dependencies using Apache Airflow, transforming raw data into optimized PostgreSQL relational schemas via CTEs and indexed queries. |

Real-Time Streaming & Distributed Infrastructure

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|-------------------|---|
| • Project | - E-Commerce Financial Analytics Pipeline |
| • Event Streaming | - Engineered an end-to-end Kafka event streaming platform using PySpark Structured Streaming with watermarking to manage late-arriving metrics. |
| • Analytics Layer | - Implemented a containerized, multi-tier dbt model architecture from raw staging to final presentation marts with built-in data tests. |

End-to-End Predictive ML System

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| • Project | - Credit Default Prediction with Explainability |
| • Management & Optimization | - Built complete batch ingestion and ML system, tracking parallel modeling algorithms and hyperparameter tuning via MLflow. |
| • Deployment & Scaling | - Containerized the pipeline using Docker to minimize image sizes, deploying a high-availability API on Kubernetes with zero-downtime rolling updates. |

GOVERNANCE, RISK MANAGEMENT & COMPLIANCE

- | | |
|--------------------|---|
| • Projects | - Real-Time Financial Transaction Monitoring Platform |
| | - 3D Printing Operations Intelligence Platform |
| • Data Privacy | - Programmed algorithmic PII masking/anonymization loops compliant with Egyptian PDPL (Law 151/2020) and GDPR. |
| • Model Validation | - Audited algorithmic bias and prevented data leakage using temporal validation splits, SHAP values, and demographic parity checks. |

